

UNIVERSITY OF CALIFORNIA

Los Angeles

Empirical Study of the Effectiveness and Optimization of Technical Indicators in Stock
Markets

A thesis submitted in partial satisfaction
of the requirements for the degree
Master of Applied Statistics and Data Science

by

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PREVIEW

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ABSTRACT OF THE THESIS

Empirical Study of the Effectiveness and Optimization of Technical Indicators in Stock Markets

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Professor Ying Nian Wu, Chair

This paper conducts a comprehensive empirical study on technical indicators for stock market prediction using 90+ years of S&P 500 data. The research evaluates 70 indicators across four categories using advanced machine learning models (LSTM, XGBoost, Random Forest, SVR, LR), identifying 35 key indicators through Principal Component Analysis. Using 60-day rolling window backtesting, LSTM models achieve superior performance with MAE of 0.014 and MAPE of 0.008, while HLC3, OBV, TEMA, and Bollinger Bands consistently outperform others. The study's key innovation—dynamic parameter optimization using Bayesian techniques—enhances predictive accuracy by 15-25% over traditional static methods. This research establishes a replicable framework for indicator optimization, demonstrates machine learning superiority in financial forecasting, and provides practical guidance for improving algorithmic trading strategies.

The thesis of Pei Guo is approved.

Xiaowu Dai

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Ying Nian Wu, Committee Chair

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